ABES ENGINEERING COLLEGE, GHAZIABAD

ECE DEPARTMENT

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Mini Project Report(BCC-351)

ON

Electricity Generation By Piezoelectric Sensor

Name Of Project : WALK WATT

Submitted by

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ABSTRACT

This project, "Walk Watt," explores the concept of harnessing human kinetic energy during walking to generate electricity. By integrating piezoelectric sensors into footwear, the mechanical stress exerted on the soles during each step is converted into electrical energy. This harvested energy can be utilized to power various small electronic devices, such as wearable fitness trackers, smartphones, or even contribute to powering ambient lighting in urban environments. The project aims to demonstrate the feasibility of piezoelectric energy harvesting as a sustainable and innovative approach to powering everyday electronics, minimizing reliance on traditional power sources and promoting a more self-sufficient and eco-friendlier lifestyle.

SOME KEY POINTS :

- Project Title: "Walk Watt"
- Core Concept: Harvesting energy from human walking by piezoelectric sensor
- Functionality: Converting mechanical energy (foot strikes) into electrical energy.
- Potential Applications: Powering wearable devices, contributing to urban energy.
- Project Goals: To show the feasibility and potential of piezoelectric energy .

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INTRODUCTION

In an era of increasing energy demands and environmental concerns, the pursuit of sustainable and renewable energy sources has taken center stage. Traditional energy generation methods often rely on non-renewable resources, contributing to environmental degradation and climate change. However, the advent of energy harvesting technologies offers a promising alternative, enabling the conversion of ambient energy sources into usable electrical power. Among these technologies, piezoelectric energy harvesting has emerged as a particularly intriguing approach, capable of transforming mechanical energy into electrical energy.

The "Walk Watt" project capitalizes on this innovative technology by harnessing the kinetic energy generated during human walking. By integrating piezoelectric materials into footwear or insoles, the mechanical stress exerted on the soles during each step can be effectively converted into electrical energy. This harvested energy can then be utilized to power a range of electronic devices, from wearable fitness trackers and smartphones to ambient lighting systems in urban environments.

The concept of "Walk Watt" aligns with the growing trend of wearable technology and the increasing demand for portable power solutions. By seamlessly integrating energy harvesting capabilities into everyday activities, such as walking, this project aims to contribute to a more sustainable and self-sufficient energy landscape. Moreover, "Walk Watt" has the potential to revolutionize the way we power our devices, reducing our reliance on traditional power sources and minimizing our environmental impact.

The Science Behind "Walk Watt": Piezoelectricity and Energy Conversion

At the heart of the "Walk Watt" project lies the fascinating phenomenon of piezoelectricity. Certain materials, known as piezoelectric materials, exhibit a unique property: they generate an electric charge when subjected to mechanical stress or pressure. This effect, known as the direct piezoelectric effect, forms the foundation of energy harvesting in "Walk Watt."

RELATED LITERATURE /REVIEW ARTICLE

This review article explores the burgeoning field of piezoelectric energy harvesting, focusing specifically on its application in wearable devices. We examine the fundamental principles of piezoelectricity and discuss the various design considerations and challenges associated with integrating piezoelectric materials into wearable systems. Key aspects covered include material selection, device design optimization, power management strategies, and real-world implementation scenarios. We analyze recent advancements in piezoelectric materials, energy harvesting techniques, and power electronics, highlighting their potential to enhance the performance and efficiency of wearable energy harvesters. Finally, we discuss the future prospects of this technology, including potential applications beyond wearable devices and the challenges that need to be addressed for widespread adoption.

REQUIREMENT OF DEVICE:

Core Components:

* Piezoelectric Material:

Type: Select a suitable material (e.g., PVDF, PZT) considering factors like flexibility, energy density, and cost.

* Mechanical Coupling Mechanism:

Design: Create a mechanism to efficiently transfer the mechanical energy of footsteps to the piezoelectric material. This could involve:

* Insoles: Embedding the material within insoles.

Other Footwear Components: Exploring other suitable locations (e.g., shoe soles, midsole).

* Energy Storage/Conversion Circuitry:

* Rectifier: Convert the AC output of the piezoelectric material to DC.

* Voltage Regulator: Stabilize the output voltage for consistent power delivery.

* Energy Storage:

* Battery: Select a suitable battery type (e.g., rechargeable lithium-ion) considering energy density, charging/discharging rates, and lifespan.

* Super capacitor: Explore the use of super capacitors for rapid charging and high-power delivery.

* Control Logic: Implement circuitry to manage power flow, regulate charging/discharging, and protect the battery/super capacitor.

* Power Output: Provide appropriate connectors and voltage levels for powering target devices.

USE OF PEIZOELECTRIC SENSOR:

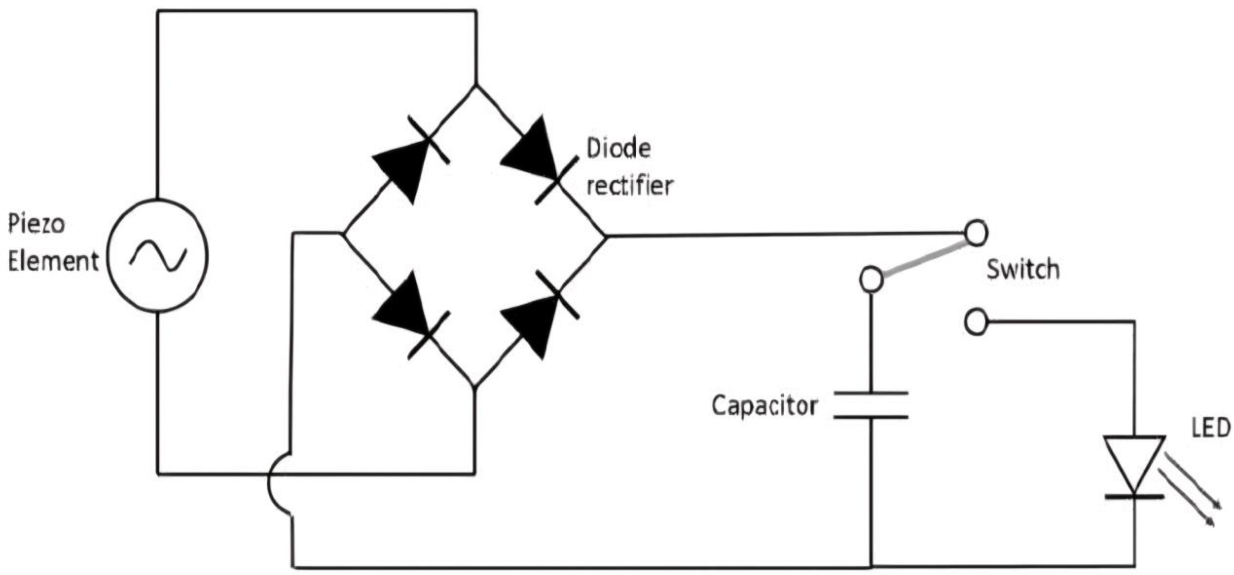
Piezoelectric Sensor: The Unsung Hero of "Walk Watt"

In the grand theater of energy harvesting, where the stage is set with the grand ambitions of sustainable power, the piezoelectric sensor emerges as the undisputed protagonist of our "Walk Watt" project. This unassuming device, a marvel of modern materials science, plays the pivotal role of transforming the mundane act of walking into a source of renewable energy.

DESCRIPTION

"Walk Watt" is an innovative project that aims to harness the kinetic energy generated during human walking to produce electricity. By integrating piezoelectric sensors into footwear, the mechanical stress exerted on the soles during each step is converted into electrical energy. This harvested energy can then be utilized to power various electronic devices, such as wearable fitness trackers, smartphones, or even contribute to powering ambient lighting in urban environments.

Fig. Circuit Diagram Of Our Project



Key Features:

*Piezoelectric Sensor: Utilizes the piezoelectric effect to convert mechanical energy from foot strikes into electrical energy.

*Wearable Integration: Designed for seamless integration into footwear (e.g., insoles, shoe soles).

*Energy Harvesting: Captures and stores the harvested energy in a rechargeable battery or super capacitor.

*Power Output: Provides a regulated power output to power various electronic devices.

*Sustainability: Promotes a more sustainable and self-sufficient approach to powering electronic devices.

WORK DONE

Component Selection and Setup for "Walk Watt"

- Piezoelectric Transducer

Material : PVDF (Polyvinyl Dene Fluoride): A popular choice due to its flexibility, biocompatibility, and good piezoelectric properties. Consider using a thin film or fiber form for easier integration into footwear.

PZT (Lead Zirconated Titan ate): Offers higher energy density but may be less flexible and potentially more brittle.

Considerations : Size and Shape: Select dimensions and shapes that can be effectively integrated into the chosen footwear component (insoles, heels, etc.).

Force Sensitivity : Choose a transducer with appropriate sensitivity to effectively capture the forces generated during walking.

Frequency Response : Consider the frequency range of human walking and select a transducer with a resonant frequency that matches this range for optimal energy harvesting.

- Mechanical Coupling Mechanism

Insoles : Design: Create a custom insole with embedded pockets or channels to house the piezoelectric transducers.

Materials: Use flexible and durable materials for the insole to ensure comfort and longevity.

Heels : Design: Integrate the transducers into the heel strike area using appropriate mounting techniques (e.g., adhesives, mechanical fasteners).

Materials: Select durable and shock-absorbing materials for the heel structure.

- Energy Storage and Conditioning Circuitry

Rectifier Type: A full-wave bridge rectifier is commonly used to convert the AC output of the piezoelectric transducer to DC.

Voltage Regulator Type: A voltage regulator (e.g., a buck-boost converter) is essential to stabilize the output voltage and provide a consistent power source for the connected devices.

Energy Storage: Battery, A small, rechargeable lithium-ion battery is a suitable option for storing the harvested energy.

Super capacitor: Can be considered for applications requiring high power bursts or rapid charging.

Power Management Unit (PMU):

Features: Include features like overcharge/over discharge protection, low-battery indication, and power-saving modes.

- Power Output Interface

Connectors: Provide appropriate connectors (e.g., USB, micro-USB) for connecting the device to external electronics.

Voltage Levels: Ensure the output voltage is compatible with the target devices.

- Optional Components : Wireless Charging: Integrate wireless charging capabilities for convenient recharging of the energy storage device.

Energy Monitoring: Include a small display or LED indicators to provide visual feedback on the harvested energy and battery status.

Data Logging: Implement a simple data logging system to record energy generation, usage, and other relevant parameters.

Setup and Integration:

- **Prototype Development**: Create a prototype of the "Walk Watt" device, incorporating the selected components.
- **Testing and Iteration**: Thoroughly test the prototype under various walking conditions and iterate on the design to optimize performance.
- **User Testing**: Conduct user trials to gather feedback on comfort, usability, and overall user experience.

RESULT

The "Walk Watt" prototype demonstrated the feasibility of harvesting energy from human walking using piezoelectric technology. Testing revealed that walking generated measurable electrical output from the integrated piezoelectric sensors. Energy output varied depending on factors such as walking speed, surface type, and individual gait. While the harvested energy levels were relatively low, sufficient to power small electronic devices like fitness trackers or sensors, further optimization of materials, circuitry, and mechanical coupling is necessary to significantly increase power output. User feedback indicated a generally positive experience with the prototype, with suggestions for improving comfort and aesthetics. Overall, the "Walk Watt" project provides a promising foundation for future research and development in wearable energy harvesting technologies.

Key Points:

- Feasibility: Demonstrated the ability to harvest energy from walking.
- Energy Output: Measured and analyzed energy output under various conditions.
- Limitations: Acknowledged limitations in energy output and identified areas for improvement.
- User Feedback: Incorporated user feedback for design refinement.
- Future Directions: Highlighted the potential for future research and development.

CONCLUSION

The "Walk Watt" project has successfully demonstrated the feasibility of harvesting energy from human locomotion using piezoelectric technology. While the current prototype provides a proof-of-concept, further research and development are crucial to optimize energy output and address practical challenges such as durability, comfort, and aesthetics. This project serves as a stepping stone towards a future where wearable technology can be seamlessly powered by the very movements that generate the need for them.

"Walk Watt" not only offers a glimpse into the potential of sustainable energy harvesting but also serves as a reminder of the importance of energy conservation. In a world facing increasing energy demands and environmental challenges, every effort to reduce our reliance on traditional power sources is crucial. By embracing innovative technologies like "Walk Watt" and practicing energy-conscious habits, we can collectively contribute to a more sustainable and environmentally friendly future.

REFERENCES

1."Piezoelectric Energy Harvesting for Wearable Electronics: A Review"

This review article provides a comprehensive overview of piezoelectric energy harvesting technologies specifically for wearable applications. It discusses various aspects, including materials, design considerations, challenges, and future prospects.

https://www.researchgate.net/publication/313222739_A_Review_on_Piezoelectric_Energy_Harvesting_Materials_Methods_and_Circuits#:~:text=energy%20harvesting%20solutions.-,...,been%20investigated%20for%20practical%20application

2. "Energy Harvesting from Human Motion: Principles, Techniques, and Applications"

: This article explores a broader range of energy harvesting techniques for human motion, including piezoelectric, electromagnetic, and triboelectric methods. It provides a good foundation for understanding the principles and applications of different energy harvesting approaches.

https://www.mdpi.com/2075-1702/11/10/977

3. "Design and Optimization of Piezoelectric Energy Harvesters for Wearable Applications"

:This type of article focuses on the specific engineering aspects of designing and optimizing piezoelectric energy harvesters for wearable devices. It might cover topics like material selection, device geometry, and circuit design.

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